

# Beau Elson · LHP | Fastball HAA Analysis

14.2%

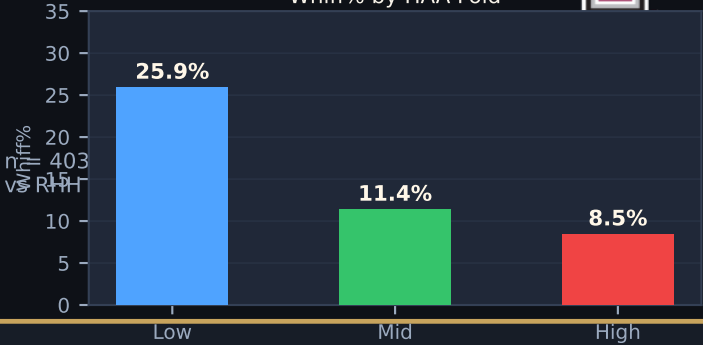
Whiff% (vs RHH)

23.3%

CSW% (vs RHH)

57.1%

Zone% (vs RHH)



**KEY FINDING**

Low HAA ( $\approx 1.08^\circ$ ) generates 25.9% Whiff% — 3 $\times$  the rate of High HAA (8.5%).

When Beau's fastball exits the hand at a lower horizontal approach angle, it arrives from the 1B side and finishes outside to right-handed hitters. That deceptive approach angle makes it extremely difficult to square up, driving the dramatic difference in swing-and-miss rate across folds.

Low HAA (25.9% Whiff), Mid HAA (11.4% Whiff), High HAA (8.5% Whiff)

HAA and PlateLocSide share a -0.90 correlation — the approach angle dictates where the ball ends up, not the other way around.

**TARGET RELEASE POINT**

Overall mean RelSide: -2.35 ft

Low-HAA fold RelSide: -2.25 ft

**Required shift:**  $\Delta = +0.10$  ft (~1.2 in toward 1B/glove side)

For a LHP, moving the release point ~1.2 inches toward the 1B / glove side naturally lowers HAA. The ball arrives from the 1B side and stays outside to RHH — the most effective approach angle.

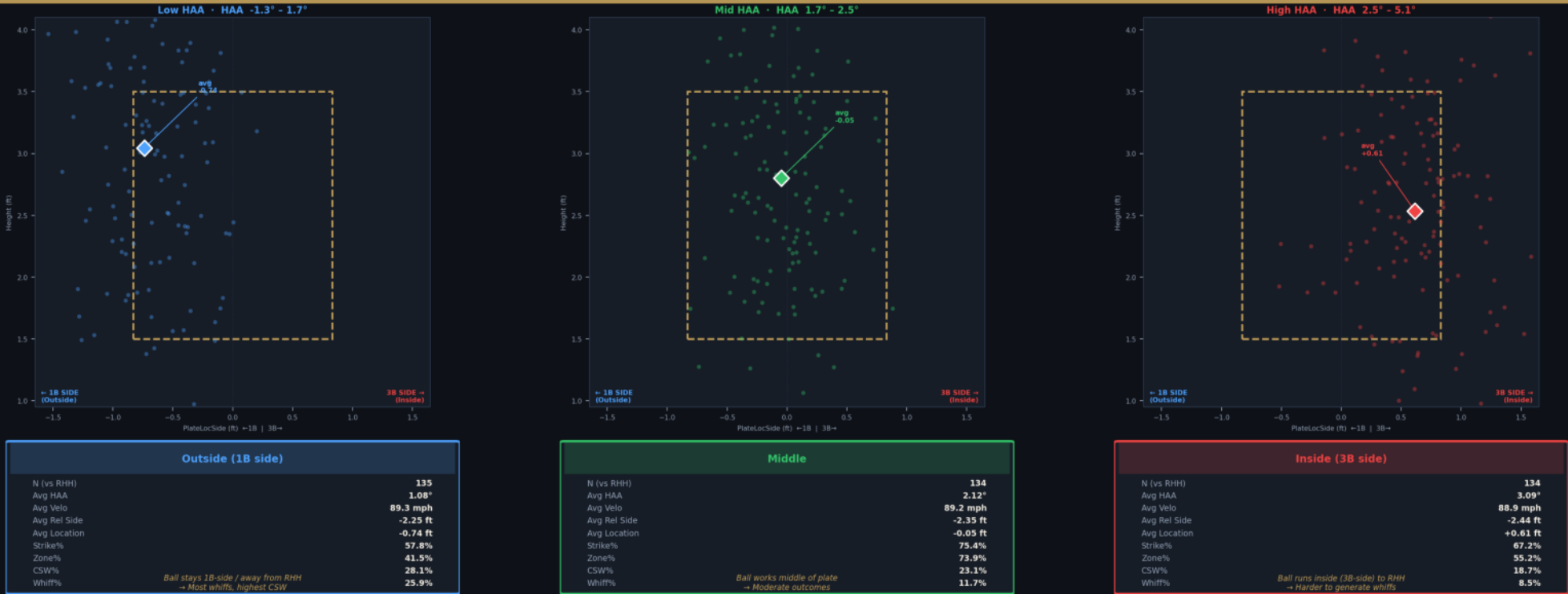
| Fold | N   | Avg HAA | Avg RelSide | Avg PlateLocSide | Swing% | Whiff% | CSW%  | Zone% |
|------|-----|---------|-------------|------------------|--------|--------|-------|-------|
| Low  | 135 | 1.08°   | -2.25 ft    | -0.74 ft         | 40.0%  | 25.9%  | 28.1% | 41.5% |
| Mid  | 134 | 2.12°   | -2.35 ft    | -0.05 ft         | 59.0%  | 11.4%  | 23.1% | 73.9% |
| High | 134 | 3.09°   | -2.44 ft    | +0.61 ft         | 53.0%  | 8.5%   | 18.7% | 56.0% |

Pair with location discipline or pitch-to-contact mix.



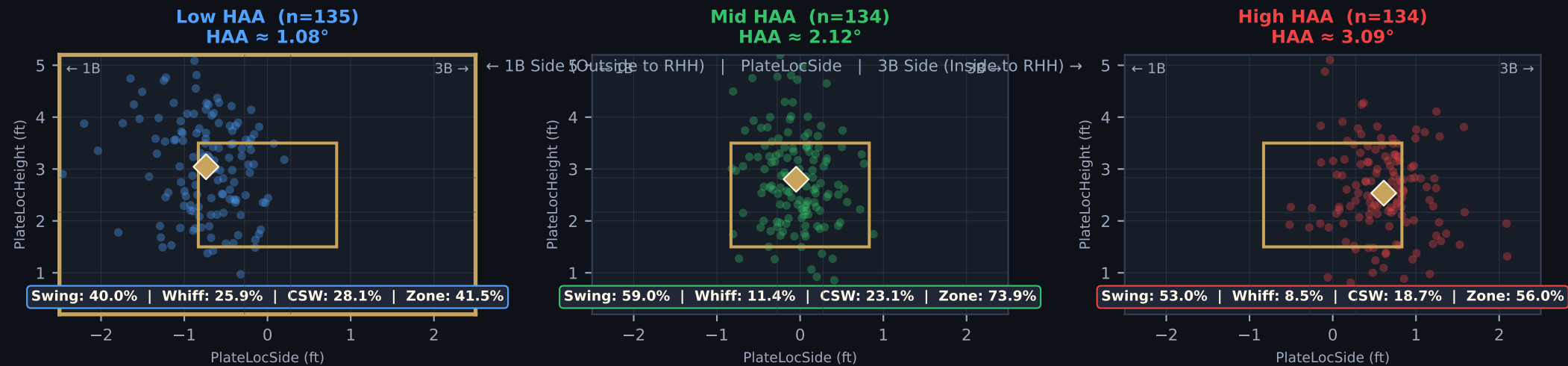
Elson, Beau (LHP) — Fastball HAA Fold Analysis · vs RHH Only

Fastballs split into 3 equal HAA terciles · Positive HAA = ball arriving from 3B side (inside to RHH) · Negative/low HAA = ball arrives from 1B side (outside to RHH) · Low HAA most effective



Three HAA tercile folds (Low / Mid / High) vs RHH | Low fold: HAA ≈ 1.08° → 25.9% Whiff% · Mid fold: HAA ≈ 2.12° → 11.4% Whiff% · High fold: HAA ≈ 3.09° → 8.5% Whiff%

# Beau Elson · Zone Location by HAA Fold (vs RHH)



Fold Comparison Table (Fastball vs RHH)

| Fold | N   | Avg HAA | RelSide (ft) | PlateLocSide (ft) | Swing% | Whiff% | CSW%  | Zone% |           |
|------|-----|---------|--------------|-------------------|--------|--------|-------|-------|-----------|
| Low  | 135 | 1.08°   | -2.25        | -0.74             | 40.0%  | 25.9%  | 28.1% | 41.5% | ← OPTIMAL |
| Mid  | 134 | 2.12°   | -2.35        | -0.05             | 59.0%  | 11.4%  | 23.1% | 73.9% |           |
| High | 134 | 3.09°   | -2.44        | +0.61             | 53.0%  | 8.5%   | 18.7% | 56.0% |           |



RUBBER / RELEASE RECOMMENDATION

Current mean RelSide: -2.35 ft

Target RelSide (Low HAA fold): -2.25 ft

Required shift: Δ = +0.10 ft ≈ 1.2 in toward 1B/glove side

Expected outcome: Whiff% from ~14.2% (overall) into 25.9% range

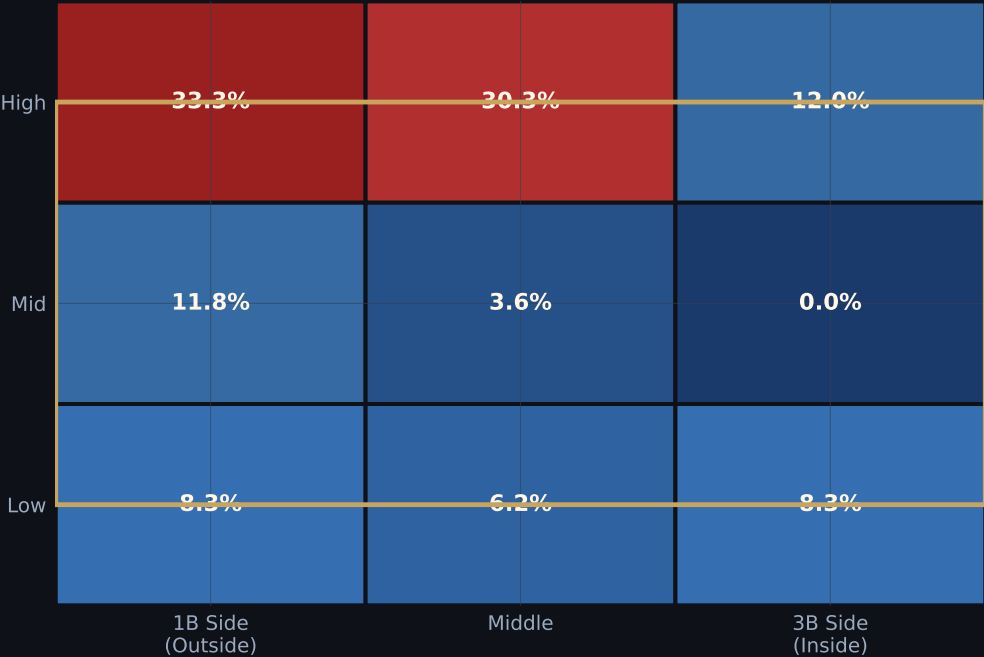
Mechanism:

For a LHP, moving release toward the 1B/glove side lowers HAA → ball arrives from 1B side → stays outside vs RHH → most effective approach angle.

Guardrail:

Zone% drops to 41.5% in Low-HAA fold (vs 73.9% Mid). Must pair with pitch-to-contact mix or sharp location discipline to maintain acceptable in-zone rates.

9-Zone Whiff% (All vs-RHH Fastballs)



STRATEGIC READ PER FOLD

LOW HAA (≈1.08°) — OPTIMAL

- Ball arrives from 1B side, finishing outside to RHH.
- 25.9% Whiff% — 3× the rate of High-HAA zone.
- Zone% is 41.5%; effective even when missing slightly off plate.
- Primary use case: put-away pitch / two-strike count; must spot carefully.

MID HAA (≈2.12°) — NEUTRAL

- Ball centers in the zone (PlateLocSide ≈ -0.05 ft). High Zone% (73.9%).
- Moderate swing-and-miss (11.4%). Good for early counts / strike-one sequencing.
- Effective when establishing zone presence before going to Low-HAA for whiff.

HIGH HAA (≈3.09°) — AVOID FOR WHIFFS

- Ball runs inside to RHH (PlateLocSide +0.61 ft). Harder to miss.
- Only 8.5% Whiff% — hitters can track and barrel the pitch.
- Zone% (56%) is moderate. Risk of hard contact on middle-in fastballs.
- Use sparingly; best only as a complementary location for pitch mix variety.